POSITIVE HELSON MATRICES WITHOUT BOUNDED SYMBOLS

ØISTEIN SØVIK

Abstract. Perfekt and Pushnitski showed that a non-negative Helson matrix is bounded exactly when its moment measure is a Carleson measure for the Hardy space $H^2(\mathbb{T}^\infty)$, and asked whether such measures on $(-1,1)^\infty$ are also Carleson measures for $H^1(\mathbb{T}^\infty)$. A positive answer would give every bounded non-negative Helson matrix a bounded symbol. We show that the answer is negative. There is a countably atomic positive measure on $(-1,1)^\infty \cap \ell^2$, supported on sign changes of the curve $(2^{-s}, 3^{-s}, 5^{-s}, \dots)$ with $s > 1/2$, whose Helson matrix is bounded and non-negative, but whose moment functional is unbounded on H^1. Hence the matrix has no bounded symbol. The proof rests on a gain for two-point measures in many small coordinates, which we compute in a Gaussian limit. We also give a computer-assisted two-point measure in three variables for which the moment functional on H^1 has norm strictly larger than the norm of the Helson matrix.

1. Introduction

Let $\mathbb{T}^\infty$ be the infinite torus with its normalized Haar measure m. For $p = 1, 2$ the Hardy space $H^p(\mathbb{T}^\infty)$ is the closure in $L^p(\mathbb{T}^\infty)$ of the analytic polynomials, that is, of the finite sums $F(z) = \sum_\kappa a_\kappa z^\kappa$ over multi-indices κ with finitely many nonzero entries, and

$$\|F\|_p = \left(\int_{\mathbb{T}^\infty} |F(z)|^p \, \mathrm{d}m(z) \right)^{1/p}.$$

Let $p_1 < p_2 < \cdots$ be the primes. If $n = \prod_j p_j^{\kappa_j}$, write $\kappa(n) = (\kappa_1, \kappa_2, \dots)$. A *Helson matrix* is a matrix of the form $M(\alpha) = (\alpha(mn))_{m,n\geq 1}$. Under the Bohr lift $n \mapsto z^{\kappa(n)}$ it becomes a Hankel form on $H^2(\mathbb{T}^\infty)$: if $f = \sum_n a_n z^{\kappa(n)}$ and $g = \sum_n b_n z^{\kappa(n)}$ are polynomials, then

$$\sum_{m,n\geq 1} \alpha(mn) a_m b_n = \Lambda_\alpha(fg), \qquad \text{where} \qquad \Lambda_\alpha\Big(\sum_k h_k z^{\kappa(k)} \Big) = \sum_k h_k \alpha(k).$$

Date: October 13, 2026.

2020 *Mathematics Subject Classification.* Primary 47B35; Secondary 30B50, 42B30.

Key words and phrases. Helson matrices, Nehari's theorem, Carleson measures, Hardy spaces of Dirichlet series, infinite polydisc.

A function $\psi \in L^\infty(\mathbb{T}^\infty)$ is a *bounded symbol* for $M(\alpha)$ if $\langle z^{\kappa(n)}, \psi\rangle = \alpha(n)$ for every $n \geq 1$. In that case $|\Lambda_\alpha(fg)| \leq \|\psi\|_\infty \|f\|_2 \|g\|_2$, so $M(\alpha)$ is bounded on ℓ^2.

By Nehari's theorem [5], the converse holds in one variable. Helson studied the multiplicative analogue [3, 4], and proved that every Helson matrix in the Hilbert–Schmidt class has a bounded symbol [3, Section 2] (see also [2, Section 6]). Ortega-Cerdà and Seip [6], using a method proposed in Helson's last paper, showed that the constant in Nehari's theorem on $\mathbb{T}^d$ grows at least exponentially in d. Consequently there are bounded Helson matrices without bounded symbols. Brevig and Perfekt [1, Theorem 3] later found such matrices in the Schatten class S_q for every $q > (1 - \log\pi/\log 4)^{-1}$. The matrices in these constructions are not positive. Brevig, Perfekt, Seip, Siskakis and Vukotić asked whether the multiplicative Hilbert matrix $(1/(\sqrt{mn}\log mn))_{m,n\geq 2}$, which is non-negative, has a bounded symbol [2, Section 6]. In a recent preprint, Perfekt shows that it does [7, Corollary 1.2]; we do not use this result.

Perfekt and Pushnitski [8] described the non-negative Helson matrices. Under a mild growth condition, $M(\alpha) \geq 0$ if and only if α is the moment sequence

$$\alpha(n) = \int t^{\kappa(n)} \,\mathrm{d}\mu(t) \tag{1}$$

of a positive measure μ on $\mathbb{R}^\infty$ [8, Theorem 3.3]. For a finite positive measure μ on $(-1,1)^\infty$, the matrix $M(\alpha)$ is bounded if and only if μ is a Carleson measure for $H^2(\mathbb{T}^\infty)$, that is, $\int |F|^2 \,\mathrm{d}\mu \leq C\|F\|_2^2$ for every polynomial F [8, Theorem 4.7]. If μ is even a Carleson measure for $H^1(\mathbb{T}^\infty)$, then $\Lambda_\alpha(F) = \int F \,\mathrm{d}\mu$ is bounded on H^1, and the Hahn–Banach theorem gives a bounded symbol. Perfekt and Pushnitski wrote, in [8, Section 4.4, after Remark 4.11]: "In general there are Carleson measures for $H^2(\mathbb{T}^\infty)$ which are not Carleson measures for $H^1(\mathbb{T}^\infty)$, but we do not know if these can be chosen to be concentrated on $(-1,1)^\infty$". They referred to [2, pp. 430–431] for a related question. We state their question as follows.

Problem 1 (Perfekt and Pushnitski)**.** *Is there a Carleson measure for $H^2(\mathbb{T}^\infty)$, concentrated on $(-1,1)^\infty$, which is not a Carleson measure for $H^1(\mathbb{T}^\infty)$?*

A negative answer would give a bounded symbol to every bounded non-negative Helson matrix. Moreover, every bounded non-negative Helson matrix has a moment measure on $(-1,1)^\infty$ (see Section 2). So the question whether every bounded non-negative Helson matrix has a bounded symbol, which we call the *positive Nehari question*, may be stated in two forms: either the functional $F \mapsto \int F \,\mathrm{d}\mu$ is bounded on H^1 for every Carleson measure μ for $H^2(\mathbb{T}^\infty)$ on $(-1,1)^\infty$, or, in the stronger form suggested by Problem 1, every such μ is a Carleson measure for $H^1(\mathbb{T}^\infty)$.

The main purpose of the present paper is to show that the answer to both forms is negative. For $s > 1/2$ let $\gamma(s) = (p_j^{-s})_{j\geq 1}$, a point of $(0,1)^\infty \cap \ell^2$.

For a finite positive measure μ on $(-1,1)^\infty$ let $C_2(\mu)$ denote the smallest constant C in the inequality $\int |F|^2 \,\mathrm{d}\mu \le C\|F\|_2^2$, and put

$$D(\mu) = \sup_{F \neq 0} \frac{\left|\int F \,\mathrm{d}\mu\right|}{\|F\|_1},$$

where F ranges over the polynomials.

Theorem 1. *There is a countably atomic positive measure μ of total mass at most 1 with $C_2(\mu) \le 1$ and $D(\mu) = \infty$. Every atom of μ is a point $\varepsilon\gamma(s) = (\varepsilon_j p_j^{-s})_{j\ge 1}$ with $s > 1/2$ and $\varepsilon \in \{-1,1\}^{\mathbb{N}}$ having only finitely many entries equal to -1.*

In particular, μ is a Carleson measure for $H^2(\mathbb{T}^\infty)$, concentrated on $(-1,1)^\infty \cap \ell^2$, which is not a Carleson measure for $H^1(\mathbb{T}^\infty)$. This answers Problem 1 in the affirmative. The Helson matrix $M(\alpha)$ given by (1) is bounded and non-negative, but has no bounded symbol, since a bounded symbol would make $D(\mu)$ finite (see Lemma 1).

The atoms of μ lie on countably many copies $\varepsilon\gamma$ of the curve γ, each rotated by a sign pattern. The entries of the multiplicative Hilbert matrix are $\alpha(mn)$ with $\alpha(n) = 1/(\sqrt{n}\log n) = \int_{1/2}^\infty \gamma(s)^{\kappa(n)}\,\mathrm{d}s$, so its moment measure is the image of Lebesgue measure on $(1/2,\infty)$ under γ, and Perfekt's bounded symbol for it comes from an H^1 estimate on the curve [7, Theorem 1.1]. Theorem 1 shows that no such estimate holds for countable sums of measures on rotated copies of the curve.

Our second result is an explicit two-point measure in three variables. Let $r = 4/5$, $x = (r,r,r)$, and let $\sigma = (\delta_x + \delta_{-x})/2$, viewed as a measure on $(-1,1)^\infty$ whose atoms have all coordinates beyond the third equal to 0.

Theorem 2. *The measure σ satisfies*

$$C_2(\sigma) = A = \frac{544140625}{50243409} \qquad \text{and} \qquad D(\sigma) > A.$$

The proof of Theorem 2 is computer-assisted. Since $C_2(\sigma) \le D(\sigma)$ for every measure on real points (see Section 6), Theorem 2 shows that this inequality can be strict already for two atoms. Tensor powers of σ on disjoint blocks of coordinates then give a second counterexample (Corollary 1), with finitely supported atoms.

To the best of our knowledge, Theorems 1 and 2 are new. Let us briefly describe our approach. The basic object is the two-point measure $\sigma_x = (\delta_x + \delta_{-x})/2$ for a vector x with entries in $(0,1)$. Its H^2 extremal is the sum U_x of the reproducing kernels at the two atoms, and U_x^2 attains the ratio $C_2(\sigma_x)$ in the definition of $D(\sigma_x)$. So the natural test function for $D(\sigma_x)$ is the square of the H^2 extremal, and it only gives $D(\sigma_x) \ge C_2(\sigma_x)$. To do better we subtract a small constant $\delta > 0$. This lowers $\int U_x^2 \,\mathrm{d}\sigma_x$ by δ, but lowers $\|U_x^2\|_1$ by δ times the mean of the phase $\mathrm{Re}(\overline{U_x}/U_x)$, up to $o(\delta)$.

Hence $D(\sigma_x) > C_2(\sigma_x)$ as soon as

$$C_2(\sigma_x) \int_{\mathbb{T}^\infty} \operatorname{Re} \frac{\overline{U_x}}{U_x} \, \mathrm{d}m > 1.$$

If U_x had no zeros in a neighbourhood of the closed polydisc, then the function $C_2(\sigma_x)\overline{U_x}/U_x$ would be a bounded symbol for σ_x, and the left-hand side would be exactly 1 (see Section 6). So the inequality can only hold when U_x has zeros there, and it measures how far the phase of U_x is from being a symbol.

For Theorem 1 we take x with many small coordinates and $\sum_j x_j^2 = 3$. Then $W_x = \sum_j x_j z_j$ is a sum of many small independent Steinhaus variables on $\mathbb{T}^\infty$, so by the central limit theorem it is nearly a complex Gaussian variable W. Since U_x is close to $e^{W_x} + e^{-W_x}$, it converges in law to $2\cosh W$, whose zeros $W \in i\pi(\mathbb{Z} + 1/2)$ are seen by the Gaussian. In this limit $C_2(\sigma_x) \to \cosh 3 > 5$ and the phase integral exceeds $5/24$, by a computation which can be done by hand. This gives a gain $\rho > 1$, uniform over all blocks of small coordinates. Gains multiply over disjoint blocks of coordinates: C_2 is submultiplicative for products of measures, while products of test functions give ρ^m for m blocks. The point $\gamma(s)$, for s close to $1/2$, has $\sum_j p_j^{-2s}$ large but all late coordinates small, so it splits into many such blocks, and flipping the sign of each block gives the atoms $\varepsilon\gamma(s)$. Finally we sum the normalized product measures over a sequence of m. Our test functions are positive on the real cube, so the other summands only add to $\int F \, \mathrm{d}\mu$, and nothing cancels.

For Theorem 2 the coordinates are few and not small, and the phase integral is a three-dimensional integral, which we bound from below in interval arithmetic.

Organization. The present paper is organized into five additional sections. In Section 2 we collect the facts on Helson matrices and Carleson measures that we need. In Section 3 we prove the uniform gain for blocks of small coordinates. Section 4 contains the proof of Theorem 1. In Section 5 we prove Theorem 2 and its corollary. Section 6 explains why the extremal must have zeros and records a question.

2. Preliminaries

Throughout, measures on $(-1,1)^\infty$ are finite positive Borel measures for the product topology. A polynomial depends on finitely many variables, so it can be evaluated at every point of $(-1,1)^\infty$. For such a measure μ and $r = 1, 2$ let

$$C_r(\mu) = \sup_{F \neq 0} \frac{\int |F|^r \, \mathrm{d}\mu}{\|F\|_r^r},$$

where F ranges over the polynomials. Then μ is a Carleson measure for $H^r(\mathbb{T}^\infty)$ if and only if $C_r(\mu) < \infty$. By the triangle inequality, $D(\mu) \leq C_1(\mu)$.

Testing with $F = 1$ shows that the total mass of μ is at most $C_2(\mu)$. If $\mu = \sum_k \mu_k$, then $C_2(\mu) \le \sum_k C_2(\mu_k)$.

We will use test functions which are not polynomials. Suppose that G is holomorphic in a neighbourhood of the closed polydisc $\overline{\mathbb{D}}^d$. Then the partial sums, by total degree, of its Taylor series converge to G uniformly on $\overline{\mathbb{D}}^d$. If G has real Taylor coefficients and $G > 0$ on $[-1,1]^d$, then the partial sums have real coefficients and are positive on $[-1,1]^d$ from some degree on. Hence the suprema defining $C_r(\mu)$ and $D(\mu)$ do not change if such functions G are admitted.

For $y \in (-1,1)^d$ the reproducing kernel of $H^2(\mathbb{T}^d)$ at y is

$$K_y(z) = \prod_{j=1}^{d} \frac{1}{1 - y_j z_j}, \qquad F(y) = \langle F, K_y \rangle, \qquad \|K_y\|_2^2 = \prod_{j=1}^{d} \frac{1}{1 - y_j^2}.$$

Note that $K_y(y') = \langle K_y, K_{y'} \rangle$ for $y, y' \in (-1,1)^d$.

Symbols and moment measures. The first lemma is the standard duality between bounded symbols and H^1; see also [2, Section 6].

Lemma 1. *A Helson matrix $M(\alpha)$ has a bounded symbol if and only if Λ_α is bounded on the polynomials in the norm of $H^1(\mathbb{T}^\infty)$.*

Proof. If ψ is a bounded symbol, then $\Lambda_\alpha(H) = \langle H, \psi \rangle$ for every polynomial H, so $|\Lambda_\alpha(H)| \le \|\psi\|_\infty \|H\|_1$. Conversely, if Λ_α is bounded, it extends by the Hahn–Banach theorem to a bounded functional on $L^1(\mathbb{T}^\infty)$ with the same norm. Such a functional is $H \mapsto \langle H, \psi \rangle$ for some $\psi \in L^\infty(\mathbb{T}^\infty)$, and $\langle z^{\kappa(n)}, \psi \rangle = \Lambda_\alpha(z^{\kappa(n)}) = \alpha(n)$. □

The next lemma is the part of [8, Theorem 4.7] that we need, with the precise norm.

Lemma 2. *Suppose that μ is a finite positive measure on $(-1,1)^\infty$, and define α by (1). Then $M(\alpha) \ge 0$, $\|M(\alpha)\| = C_2(\mu)$, and $\Lambda_\alpha(H) = \int H \, \mathrm{d}\mu$ for every polynomial H. In particular, $M(\alpha)$ has a bounded symbol if and only if $D(\mu) < \infty$.*

Proof. Since $|t^\kappa| \le 1$ on $(-1,1)^\infty$, the moments are finite, and $\Lambda_\alpha(H) = \int H \, \mathrm{d}\mu$ by linearity. Let (a_n) be finitely supported, and let $F = \sum_n a_n z^{\kappa(n)}$. Since $t^{\kappa(n)}$ is real,

$$\sum_{m,n \ge 1} \alpha(mn) a_m \overline{a_n} = \int F(t) \overline{F(t)} \, \mathrm{d}\mu(t) = \int |F|^2 \, \mathrm{d}\mu.$$

Hence $M(\alpha) \ge 0$, and its norm is the supremum of this quadratic form over $\|F\|_2 = \|a\|_{\ell^2} = 1$, which is $C_2(\mu)$. The final claim follows from Lemma 1. □

Conversely, suppose that $M(\alpha)$ is bounded and non-negative. Then $|\alpha(n)| \le \|M(\alpha)\|$, so [8, Theorem 3.3] gives a positive measure μ on $\mathbb{R}^\infty$ with (1). Fix j. Then $\int t_j^{2k} \, \mathrm{d}\mu = \alpha(p_j^{2k}) \le \|M(\alpha)\|$ for every k, so $\mu(|t_j| >$

$1) = 0$. The first column of $M(\alpha)$ is in ℓ^2, so $\alpha(p_j^{2k}) \to 0$ as $k \to \infty$. Since $\int t_j^{2k} \, \mathrm{d}\mu \geq \mu(|t_j| = 1)$, we get $\mu(|t_j| = 1) = 0$. Hence μ is concentrated on $(-1,1)^\infty$, as also noted in [8, Section 4.4]. This justifies the two forms of the positive Nehari question stated in the introduction.

Products and two-point measures. We will build measures as products over disjoint sets of coordinates. Let $\mathbb{N} = I_1 \cup I_2$ be a partition, and let μ_i be a finite positive measure on $(-1,1)^{I_i}$. We define $C_2(\mu_i)$ and $D(\mu_i)$ with polynomials in the variables $(z_j)_{j \in I_i}$, and identify $\mu_1 \otimes \mu_2$ with a measure on $(-1,1)^\infty$.

Lemma 3. *Suppose that μ_1 and μ_2 are as above. Then $C_2(\mu_1 \otimes \mu_2) \leq C_2(\mu_1) C_2(\mu_2)$. If F_i is a polynomial in the variables indexed by I_i for $i = 1, 2$, then*

$$\|F_1 F_2\|_1 = \|F_1\|_1 \|F_2\|_1, \qquad \int F_1 F_2 \, \mathrm{d}(\mu_1 \otimes \mu_2) = \int F_1 \, \mathrm{d}\mu_1 \int F_2 \, \mathrm{d}\mu_2.$$

Proof. The identities hold by Fubini's theorem, since m is the product of the Haar measures on $\mathbb{T}^{I_1}$ and $\mathbb{T}^{I_2}$. Let F be a polynomial and write $z = (u, v)$ with $u \in \mathbb{T}^{I_1}$ and $v \in \mathbb{T}^{I_2}$. For fixed $t \in (-1,1)^{I_1}$, the function $v \mapsto F(t, v)$ is a polynomial, so

$$\int |F(t,w)|^2 \, \mathrm{d}\mu_2(w) \leq C_2(\mu_2) \int_{\mathbb{T}^{I_2}} |F(t,v)|^2 \, \mathrm{d}m(v).$$

We integrate in t against μ_1, use Fubini's theorem, and apply the same inequality to the polynomial $u \mapsto F(u, v)$ for each fixed v. This gives $\int |F|^2 \, \mathrm{d}(\mu_1 \otimes \mu_2) \leq C_2(\mu_1) C_2(\mu_2) \|F\|_2^2$. □

For a vector $x \in (0,1)^d$ we write $\sigma_x = (\delta_x + \delta_{-x})/2$ and

$$A_x = \frac{1}{2}\Big(\prod_{j=1}^d \frac{1}{1 - x_j^2} + \prod_{j=1}^d \frac{1}{1 + x_j^2} \Big), \qquad U_x(z) = \prod_{j=1}^d \frac{1}{1 - x_j z_j} + \prod_{j=1}^d \frac{1}{1 + x_j z_j}.$$

The next lemma computes the H^2 constant of σ_x and of a point mass, and records the test functions that we will perturb.

Lemma 4. *Suppose that $x \in (0,1)^d$. Then $C_2(\sigma_x) = A_x$, $U_x(x) = U_x(-x) = 2A_x$ and $\|U_x\|_2^2 = 4A_x$. In particular,*

$$\int U_x^2 \, \mathrm{d}\sigma_x = 4A_x^2 = A_x \|U_x^2\|_1. \tag{2}$$

Moreover, $U_x \geq 2$ on $[-1,1]^d$. Suppose that $I \subseteq \mathbb{N}$ and that $y = (y_j)_{j \in I}$ is a point of $(-1,1)^I \cap \ell^2$, and put $B = \prod_{j \in I} (1 - y_j^2)^{-1}$. Then $C_2(\delta_y) \leq B$, and for every $\theta > 0$ there is a polynomial G in the variables indexed by a finite set $J \subseteq I$, positive on $[-1,1]^J$, with $G(y) \geq (1 - \theta) B \|G\|_1$.

Proof. For a polynomial F in d variables,

$$\int |F|^2 \, \mathrm{d}\sigma_x = \frac{1}{2}\Big(|\langle F, K_x \rangle|^2 + |\langle F, K_{-x} \rangle|^2 \Big).$$

So $C_2(\sigma_x)$ is the square of the norm of the map $F \mapsto (\langle F, K_x\rangle, \langle F, K_{-x}\rangle)/\sqrt{2}$, which is the largest eigenvalue of half the Gram matrix of K_x and K_{-x}, which is

$$\frac{1}{2}\begin{pmatrix} a & b \\ b & a \end{pmatrix}, \qquad a = \prod_{j=1}^{d} \frac{1}{1-x_j^2}, \quad b = \prod_{j=1}^{d} \frac{1}{1+x_j^2}.$$

Its eigenvalues are $(a \pm b)/2$, so $C_2(\sigma_x) = (a+b)/2 = A_x$, with eigenvector $U_x = K_x + K_{-x}$. Then $U_x(\pm x) = a + b = 2A_x$ and $\|U_x\|_2^2 = 2a + 2b = 4A_x$. Since $\|U_x^2\|_1 = \|U_x\|_2^2$, this gives (2). For $t \in [-1,1]^d$ both terms of $U_x(t)$ are positive, and their product is $\prod_j (1 - x_j^2 t_j^2)^{-1} \geq 1$. By the inequality of arithmetic and geometric means, $U_x(t) \geq 2$.

If F is a polynomial in the variables indexed by a finite set J, then $|F(y)|^2 \leq \|K_{y_J}\|_2^2 \|F\|_2^2 \leq B\|F\|_2^2$, where y_J is the restriction of y to J. Hence $C_2(\delta_y) \leq B$. For the last claim let $G_J = K_{y_J}^2$. Then $\|G_J\|_1 = \|K_{y_J}\|_2^2$ and $G_J(y) = \|K_{y_J}\|_2^4$, so $G_J(y)/\|G_J\|_1 = \prod_{j\in J}(1-y_j^2)^{-1}$, which tends to B as J increases to I. Since $G_J > 0$ on $[-1,1]^J$, the approximation remark above gives the polynomial G. □

3. A uniform gain for blocks of small coordinates

By (2), the function U_x^2 attains the ratio A_x in the definition of $D(\sigma_x)$. The next lemma shows that a small perturbation of U_x^2 does strictly better, uniformly over all x with small coordinates and $\sum_j x_j^2$ close to 3.

Lemma 5. *There are $0 < \eta < 1$ and $\rho > 1$ with the following property. Suppose that $x \in (0,1)^d$ for some $d \geq 1$, that $\max_j x_j < \eta$ and that $|\sum_j x_j^2 - 3| < \eta$. Then there is a polynomial f with real coefficients, positive on $[-1,1]^d$, such that*

$$\int f \, \mathrm{d}\sigma_x \geq \rho A_x \|f\|_1.$$

Proof. The proof has three steps. We first show that U_x converges in law to $2\cosh W$ for a complex Gaussian variable W, together with the quantities in (2). We then show by hand that subtracting a small constant from $U_0^2 = 4\cosh^2 W$ gives a gain in the limit. Finally, a compactness argument transfers the gain to all x close to the limit.

We first identify the limit. Consider a sequence of vectors x with $\max_j x_j \to 0$ and $\sum_j x_j^2 \to 3$, the dimension d being allowed to vary. We regard $z \in \mathbb{T}^d$ as a random variable with distribution m, and write $W_x = \sum_j x_j z_j$. Let $W = X + iY$, where X and Y are independent real Gaussian variables with mean 0 and variance $3/2$. For $\zeta \in \mathbb{C}$, the characteristic function of a single term is

$$\mathbb{E} e^{i \operatorname{Re}(\bar{\zeta} x_j z_j)} = 1 - \frac{|\zeta|^2 x_j^2}{4} + O(|\zeta|^4 x_j^4).$$

Since $\sum_j x_j^4 \le \max_j x_j^2 \sum_j x_j^2 \to 0$, the product over j tends to $e^{-3|\zeta|^2/4} = \mathbb{E}e^{i\,\mathrm{Re}(\bar\zeta W)}$. Hence $W_x \to W$ in law. Next,

$$\log \prod_{j=1}^d \frac{1}{1 \mp x_j z_j} = \pm W_x + R_\pm, \qquad R_\pm = \sum_{j=1}^d \sum_{k\ge 2} \frac{(\pm x_j z_j)^k}{k}.$$

The monomials z_j^k are orthonormal, so

$$\mathbb{E}|R_\pm|^2 = \sum_{j=1}^d \sum_{k\ge 2} \frac{x_j^{2k}}{k^2} \le \sum_{j=1}^d \frac{x_j^4}{1-x_j^2} \to 0.$$

Hence $R_\pm \to 0$ in probability, and $U_x = e^{W_x+R_+} + e^{-W_x+R_-}$ converges in law to $U_0 = 2\cosh W$. Moreover, by orthogonality and $\sum_{k\ge 0}(k+1)^2 u^k = (1+u)/(1-u)^3$,

$$\mathbb{E}\Big|\prod_{j=1}^d \frac{1}{1 \mp x_j z_j}\Big|^4 = \prod_{j=1}^d \frac{1+x_j^2}{(1-x_j^2)^3},$$

which stays bounded. Hence $\mathbb{E}|U_x|^4$ is bounded, and $|U_x^2 - \delta|$ is uniformly integrable for every $\delta \ge 0$. We conclude that $\mathbb{E}|U_x^2 - \delta| \to \mathbb{E}|U_0^2 - \delta|$. Also $A_x \to A_0 = \cosh 3$, and by (2), $\int (U_x^2 - \delta)\,\mathrm{d}\sigma_x = 4A_x^2 - \delta \to 4A_0^2 - \delta$.

We next show that the limit admits a gain. Since W has a density, $U_0 \neq 0$ almost surely. Put

$$c_0 = \mathbb{E}\,\mathrm{Re}\,\frac{\overline{U_0}}{U_0}.$$

The function $\mathrm{Re}(\overline{U_0}/U_0)$ is unchanged if W is replaced by $-W$, so we may compute it on the event $X > 0$. There we let $a = e^{-2X} < 1$ and $\omega = e^{-2iY}$, and find that

$$\frac{\overline{\cosh W}}{\cosh W} = \omega \frac{1+a\overline{\omega}}{1+a\omega} = \frac{a+\omega}{1+a\omega} = a + (1-a^2)\sum_{k\ge 1}(-a)^{k-1}\omega^k.$$

Since X and Y are independent and $\mathbb{E}e^{-2ikY} = e^{-3k^2}$, we get

$$c_0 = \mathbb{E}\Big[a + (1-a^2)\sum_{k\ge 1}(-a)^{k-1}e^{-3k^2}\Big], \qquad a = e^{-2|X|}.$$

The series is alternating with decreasing terms, so it is nonnegative. Hence, using $e^{-u} \ge 1-u$,

$$\begin{aligned} c_0 \ge \mathbb{E}e^{-2|X|} &= \frac{2}{\sqrt{3\pi}} \int_0^\infty e^{-2v - v^2/3}\,\mathrm{d}v \\ &\ge \frac{2}{\sqrt{3\pi}} \int_0^\infty e^{-2v}\Big(1 - \frac{v^2}{3}\Big)\,\mathrm{d}v = \frac{5}{6\sqrt{3\pi}} > \frac{5}{24}, \end{aligned}$$

where the last step is $3\pi < 16$. Since $e^3 > 10$, we have $A_0 = \cosh 3 > 5$, and therefore $A_0 c_0 > 25/24 > 1$.

In words, the phase $\overline{U_0}/U_0$ consists of a positive part a and rotations $\omega^k = e^{-2ikY}$, and the rotations average out because Y has large variance.

The value 3 of $\sum_j x_j^2$ is a matter of convenience. For a larger variance v, the constant $A_0 = \cosh v$ grows exponentially, while $\mathbb{E}e^{-2|X|}$ decays only like $v^{-1/2}$.

For $w \neq 0$, the derivative of $\delta \mapsto |w - \delta|$ at $\delta = 0$ is $-\operatorname{Re}(\overline{w}/|w|)$, and the difference quotients are bounded by 1. With $w = U_0^2$ we have $\overline{w}/|w| = \overline{U_0}/U_0$. Hence dominated convergence gives

$$\frac{\mathrm{d}}{\mathrm{d}\delta}\Big(4A_0^2 - \delta - A_0\mathbb{E}|U_0^2 - \delta|\Big)\Big|_{\delta=0^+} = A_0 c_0 - 1 > 0.$$

The expression in parentheses vanishes at $\delta = 0$, since $\mathbb{E}|U_0|^2 = \lim\|U_x\|_2^2 = 4A_0$. So we may fix $0 < \delta < 1$ such that

$$\rho_0 = \frac{4A_0^2 - \delta}{A_0\mathbb{E}|U_0^2 - \delta|} > 1.$$

Let $\rho = (1 + \rho_0)/2$. We claim that there is $0 < \eta < 1$ such that

$$\int (U_x^2 - \delta)\,\mathrm{d}\sigma_x > \rho A_x \|U_x^2 - \delta\|_1 \tag{3}$$

whenever $\max_j x_j < \eta$ and $|\sum_j x_j^2 - 3| < \eta$. Otherwise there is a sequence of vectors x with $\max_j x_j \to 0$ and $\sum_j x_j^2 \to 3$ for which (3) fails. By the first part of the proof, the ratio of the two sides of (3) tends to $\rho_0/\rho > 1$ along this sequence, which is a contradiction.

Finally, $U_x^2 - \delta$ is holomorphic in a neighbourhood of $\overline{\mathbb{D}}^d$, has real Taylor coefficients, and satisfies $U_x^2 - \delta \geq 4 - \delta > 0$ on $[-1, 1]^d$ by Lemma 4. Since (3) is strict, the approximation remark of Section 2 gives a polynomial f with the stated properties. □

4. Proof of the main theorem

Armed with Lemma 5, we place many blocks of small coordinates on the curve $\gamma(s)$ close to $s = 1/2$, and multiply the gains. On each block we put the two-point measure of Lemma 5. The remaining coordinates, among them the first few, which are not small, carry a single point mass δ_y. By Lemma 4, the H^2 constant of δ_y is matched by a positive test function up to a factor 2, so this part neither helps nor hurts. By Lemma 3, the H^2 constant of the product measure is at most the product of the constants of the factors, while the product of the test functions gains ρ on every block.

Proof of Theorem 1. Let η and ρ be as in Lemma 5, and choose j_0 such that $p_j^{-1/2} < \eta/2$ for $j > j_0$. Fix $m \geq 1$. Since $\sum_j p_j^{-1}$ diverges, monotone convergence gives some $s = s_m > 1/2$ with

$$\sum_{j > j_0} p_j^{-2s} > m(3 + \eta^2/4).$$

Every term is less than $p_j^{-1} < \eta^2/4$. Running through the indices $j > j_0$ in increasing order, we may therefore choose disjoint finite sets $I_1, \ldots, I_m \subset$

$\{j > j_0\}$ with

$$3 \le \sum_{j\in I_i} p_j^{-2s} < 3 + \eta^2/4, \qquad i = 1, \dots, m.$$

Let $x^{(i)} = (p_j^{-s})_{j\in I_i}$. Then $\max x^{(i)} < \eta/2$ and $|\sum_{j\in I_i} p_j^{-2s} - 3| < \eta$, so Lemma 5 applies to each $x^{(i)}$ and gives a polynomial f_i. Let I_0 be the complement of $I_1 \cup \cdots \cup I_m$ in $\mathbb{N}$, and let $y = (p_j^{-s})_{j\in I_0}$, which lies in ℓ^2 since $s > 1/2$. Put

$$\nu_m = \delta_y \otimes \sigma_{x^{(1)}} \otimes \cdots \otimes \sigma_{x^{(m)}}, \qquad M_m = B \prod_{i=1}^{m} A_{x^{(i)}}, \qquad B = \prod_{j\in I_0} \frac{1}{1 - p_j^{-2s}}.$$

The measure ν_m is the uniform probability measure on the 2^m points $\varepsilon\gamma(s)$, where ε is constant on each I_i and equal to 1 on I_0. By Lemmas 3 and 4, $C_2(\nu_m) \le M_m$. By Lemma 4, there is a polynomial G in finitely many variables indexed by I_0, positive on the corresponding real cube, with $G(y) \ge B\|G\|_1/2$. Let $F = Gf_1 \cdots f_m$. By Lemmas 3 and 5,

$$\int F \,\mathrm{d}\nu_m = G(y) \prod_{i=1}^{m} \int f_i \,\mathrm{d}\sigma_{x^{(i)}} \ge \frac{B}{2} \prod_{i=1}^{m} \rho A_{x^{(i)}} \cdot \|G\|_1 \prod_{i=1}^{m} \|f_i\|_1 = \frac{\rho^m}{2} M_m \|F\|_1.$$

Moreover, F is positive at every point of $(-1,1)^\infty$, since each of its factors is positive on the real cube in its own variables.

Choose integers m_k with $\rho^{m_k} \ge 2 \cdot 4^k$ for $k \ge 1$, let F_k be the function F above for $m = m_k$, normalized so that $\|F_k\|_1 = 1$, and put

$$\mu = \sum_{k\ge 1} \frac{2^{-k}}{M_{m_k}} \nu_{m_k}.$$

Then $C_2(\mu) \le \sum_k 2^{-k} = 1$, so μ has total mass at most 1. Since F_k is positive on $(-1,1)^\infty$, the other terms of the sum contribute nonnegative amounts to $\int F_k \,\mathrm{d}\mu$, and so

$$\int F_k \,\mathrm{d}\mu \ge \frac{2^{-k}}{M_{m_k}} \int F_k \,\mathrm{d}\nu_{m_k} \ge 2^{-k} \frac{\rho^{m_k}}{2} \ge 2^k.$$

Hence $D(\mu) = \infty$. The atoms of μ are the points $\varepsilon\gamma(s_{m_k})$ described above, and each ε has finitely many entries equal to -1. □

Since $D(\mu) \le C_1(\mu)$, the measure μ is not a Carleson measure for $H^1(\mathbb{T}^\infty)$. By Lemma 2, the Helson matrix of μ is bounded and non-negative and has no bounded symbol.

5. A two-point measure in three variables

We now turn to the explicit measure of Theorem 2. Let $r = 4/5$, $x = (r, r, r)$, $\sigma = \sigma_x$ and $U = U_x$. By Lemma 4,

$$C_2(\sigma) = A = \frac{1}{2}\left(\left(\frac{25}{9}\right)^3 + \left(\frac{25}{41}\right)^3\right) = \frac{544140625}{50243409}.$$

Here the coordinates are few and not small, so the Gaussian limit of Section 3 is not available. Instead, we bound the phase integral directly. The idea of the computation is simple. We split $\mathbb{T}^3$ into small cells, evaluate the phase at the centre of each cell, and subtract an explicit bound for how much the phase can move within the cell. This bound is large only near the zeros of U, where we use the trivial bound -1.

Lemma 6. *The phase integral of U satisfies*

$$c = \int_{\mathbb{T}^3} \operatorname{Re} \frac{\overline{U(z)}}{U(z)} \, \mathrm{d}m(z) \geq \frac{409587562197037715}{2^{61}} > \frac{17}{100}.$$

Proof. Expanding the products, we find that

$$U(z) = \frac{2\bigl(1 + r^2(z_1z_2 + z_1z_3 + z_2z_3)\bigr)}{\prod_{j=1}^{3}(1 - r^2 z_j^2)},$$

so the zero set of U in $\mathbb{T}^3$ has measure zero. Let $L = 256$ and $h = \pi/L$, and split $\mathbb{T}^3$ into L^3 cells which are products of arcs of length $2h$, with centres $\xi = (\xi_1, \xi_2, \xi_3)$. On the cell with centre ξ we have $|1 - rz_j| \geq p_j$ and $|1 + rz_j| \geq n_j$, where

$$p_j = \max\bigl\{1 - r, |1 - r\xi_j| - rh\bigr\}, \qquad n_j = \max\bigl\{1 - r, |1 + r\xi_j| - rh\bigr\}.$$

Let $P = \prod_j p_j^{-1}$ and $N = \prod_j n_j^{-1}$. The derivative of U with respect to the angle of z_j is bounded on the cell by $r(Pp_j^{-1} + Nn_j^{-1})$. Moving one angle at a time, we find that

$$|U(z) - U(\xi)| \leq \Delta(\xi) = hr\biggl(P\sum_{j=1}^{3} p_j^{-1} + N\sum_{j=1}^{3} n_j^{-1}\biggr)$$

on the cell. Suppose that $\Delta(\xi) < |U(\xi)|$. Then the arguments of $U(z)$ and $U(\xi)$ differ by some ϕ with $|\sin\phi| \leq \Delta(\xi)/|U(\xi)|$. Since $\overline{U}/U = e^{-2i\arg U}$ and $|e^{-2i\phi} - 1| = 2|\sin\phi|$, we get

$$\operatorname{Re}\frac{\overline{U(z)}}{U(z)} \geq b(\xi) = \max\biggl\{-1, \operatorname{Re}\frac{\overline{U(\xi)}}{U(\xi)} - \frac{2\Delta(\xi)}{|U(\xi)|}\biggr\}.$$

If $\Delta(\xi) \geq |U(\xi)|$, then $b(\xi) = -1$, and the inequality is trivial. Hence $c \geq L^{-3}\sum_\xi b(\xi)$. We evaluate this sum by computer. The one-variable factors $(1 \mp r\xi_j)^{-1}$ and the bounds p_j and n_j are enclosed in Arb balls at 128-bit precision. All further arithmetic, including the evaluation of $U(\xi)$ and $b(\xi)$, uses binary64 intervals, with each operation rounded outwards. Each $b(\xi)$ is rounded down to an integer multiple of 2^{-40}, and the sum is computed exactly in integer arithmetic. The result is the stated fraction. □

Code availability. The computation in the proof of Lemma 6 is carried out by a Python script, which requires NumPy and python-flint 0.9.0. It runs over all 256^3 cells and checks the inequalities of Lemma 6 and $17/100 > 1/A$ in exact rational arithmetic. The script is `two_atom_phase.py` in the

directory `numerics/book/positive-nehari/` of the repository `https://github.com/Oisov/math-preprints`.

Proof of Theorem 2. For $\varepsilon > 0$ let $F_\varepsilon = U^2 - \varepsilon$. By (2), $\int F_\varepsilon \, \mathrm{d}\sigma = 4A^2 - \varepsilon$ and $\|F_0\|_1 = 4A$. As in the proof of Lemma 5, dominated convergence gives

$$\|F_\varepsilon\|_1 = 4A - \varepsilon c + o(\varepsilon), \qquad \varepsilon \to 0^+.$$

Hence

$$\int F_\varepsilon \, \mathrm{d}\sigma - A\|F_\varepsilon\|_1 = \varepsilon(Ac - 1) + o(\varepsilon).$$

By Lemma 6, $Ac > 17A/100 > 1$, since $1/A < 1/10$. So the left-hand side is positive for some small $\varepsilon > 0$. As F_ε is holomorphic in a neighbourhood of $\overline{\mathbb{D}}^3$, its Taylor polynomials give $D(\sigma) > A$. □

The function F_ε is at least $4 - \varepsilon$ on $[-1, 1]^3$ by Lemma 4, so the polynomial in the proof may be taken positive on $[-1, 1]^3$. This gives a second counterexample.

Corollary 1. *Let $\tau = \sigma/A$, and let τ_m be the measure $\tau^{\otimes m}$ placed on the coordinates j with $3m(m-1)/2 < j \le 3m(m+1)/2$, all other coordinates being 0. Then $\mu' = \sum_{m \ge 1} m^{-2} \tau_m$ is a countably atomic measure on $(-1, 1)^\infty \cap \ell^2$ with $C_2(\mu') \le \pi^2/6$ and $D(\mu') = \infty$.*

Proof. The point mass at 0 has $C_2(\delta_0) = 1$, so Lemma 3 gives $C_2(\tau_m) \le C_2(\tau)^m = 1$, and hence $C_2(\mu') \le \sum_m m^{-2} = \pi^2/6$. By Theorem 2 and the remark above, there is a polynomial f in three variables, positive on $[-1, 1]^3$, with $\|f\|_1 = 1$ and $\int f \, \mathrm{d}\tau = a > 1$. Let F_m be the product of m copies of f, one in each group of three coordinates of the block of τ_m. By Lemma 3, $\|F_m\|_1 = 1$ and $\int F_m \, \mathrm{d}\tau_m = a^m$. Since F_m is positive on $(-1, 1)^\infty$, we get $\int F_m \, \mathrm{d}\mu' \ge m^{-2} a^m \to \infty$. □

6. Concluding remarks

The reverse inequality. For every finite positive measure μ on $(-1, 1)^\infty$ we have $C_2(\mu) \le D(\mu)$, so that Carleson measures for $H^1(\mathbb{T}^\infty)$ are Carleson measures for $H^2(\mathbb{T}^\infty)$. Indeed, let G be a polynomial, and let $G^\sharp$ be the polynomial with conjugated coefficients. On real points $GG^\sharp = |G|^2$, and $\|GG^\sharp\|_1 \le \|G\|_2 \|G^\sharp\|_2 = \|G\|_2^2$. Hence $\int |G|^2 \, \mathrm{d}\mu \le D(\mu) \|G\|_2^2$.

Why the extremal must have zeros. The gain in Lemma 5 and Theorem 2 uses that the extremal function has zeros. Suppose that ν is a finite positive atomic measure on $(-1, 1)^d$, and that the H^2 extremal U for $C_2(\nu)$ has real coefficients and no zeros in a neighbourhood of $\overline{\mathbb{D}}^d$. The extremal satisfies $\int U \overline{g} \, \mathrm{d}\nu = C_2(\nu) \langle U, g \rangle$ for every $g \in H^2(\mathbb{T}^d)$. Let h be a polynomial and $g = (h/U)^\sharp$. Then $\overline{g(t)} = h(t)/U(t)$ for real t, and $\overline{g(z)} = h(\overline{z})/U(\overline{z})$ for $z \in \mathbb{T}^d$. Since m is invariant under $z \mapsto \overline{z}$ and $U(\overline{z}) = \overline{U(z)}$, we get

$$\int h \, \mathrm{d}\nu = C_2(\nu) \int_{\mathbb{T}^d} h \frac{\overline{U}}{U} \, \mathrm{d}m.$$

So $C_2(\nu)\overline{U}/U$ is a bounded symbol, and $D(\nu) = C_2(\nu)$. For two opposite atoms $\pm(r, \dots, r)$ in one or two variables, the numerator of U is 2 or $2(1 + r^2 z_1 z_2)$, which have no zeros in $\overline{\mathbb{D}}^d$. In three variables the numerator $2(1 + r^2(z_1z_2 + z_1z_3 + z_2z_3))$ vanishes at $z_1 = z_2 = z_3 = i/(\sqrt{3}r)$, which lies in $\mathbb{D}^3$ for $r > 1/\sqrt{3}$. This is why Theorem 2 uses three variables. A zero alone does not give $D(\nu) > C_2(\nu)$; this is the role of Lemma 6.

Scalar and absolute embeddings. Our measures satisfy $D(\mu) = C_1(\mu) = \infty$. For a single measure, the condition $D(\mu) < \infty$, which is equivalent to the existence of a bounded symbol, is a priori weaker than $C_1(\mu) < \infty$. In one variable, both conditions are equivalent to $C_2(\mu) < \infty$, by Nehari's theorem and since Carleson measures for $H^p(\mathbb{T})$ do not depend on p. We do not know whether the two conditions are equivalent on $(-1, 1)^\infty$.

Problem 2. *Suppose that μ is a finite positive measure on $(-1, 1)^\infty$ with $D(\mu) < \infty$. Is μ a Carleson measure for $H^1(\mathbb{T}^\infty)$?*

Acknowledgements. AI assistance was used in developing and checking the argument and in preparing the exposition; the author has verified the proof.

References

[1] Ole Fredrik Brevig and Karl-Mikael Perfekt, *Failure of Nehari's theorem for multiplicative Hankel forms in Schatten classes*, Studia Math. **228** (2015), no. 2, 101–108, DOI 10.4064/sm228-2-1, available at `arXiv:1409.3117`. ↑2

[2] Ole Fredrik Brevig, Karl-Mikael Perfekt, Kristian Seip, Aristomenis G. Siskakis, and Dragan Vukotić, *The multiplicative Hilbert matrix*, Adv. Math. **302** (2016), 410–432, DOI 10.1016/j.aim.2016.07.019, available at `arXiv:1411.7294`. ↑2, 5

[3] Henry Helson, *Hankel forms and sums of random variables*, Studia Math. **176** (2006), no. 1, 85–92, DOI 10.4064/sm176-1-6. ↑2

[4] ———, *Hankel forms*, Studia Math. **198** (2010), no. 1, 79–84, DOI 10.4064/sm198-1-5. ↑2

[5] Zeev Nehari, *On bounded bilinear forms*, Ann. of Math. (2) **65** (1957), 153–162, DOI 10.2307/1969670. ↑2

[6] Joaquim Ortega-Cerdà and Kristian Seip, *A lower bound in Nehari's theorem on the polydisc*, J. Anal. Math. **118** (2012), no. 1, 339–342, DOI 10.1007/s11854-012-0038-y, available at `arXiv:1107.0175`. ↑2

[7] Karl-Mikael Perfekt, *A Fejér–Riesz inequality for Dirichlet series* (2026), preprint, available at `arXiv:2609.03855`. ↑2, 3

[8] Karl-Mikael Perfekt and Alexander Pushnitski, *On Helson matrices: moment problems, non-negativity, boundedness, and finite rank*, Proc. Lond. Math. Soc. (3) **116** (2018), no. 1, 101–134, DOI 10.1112/plms.12068, available at `arXiv:1611.03772`. ↑2, 5, 6

University of Oslo, Oslo, Norway
Email address: `oistein.sovik@usit.uio.no`